

Tutorial: Analysis of Estimators, and Generalisation / Confidence Bounds

Exercise Solutions

Question 1

The table below lists three different estimators of the Bernoulli mean parameter θ , where:

$$S = \sum_{i=1}^n X_i, \quad X_i \sim \text{Bernoulli}(\theta)$$

Method	Estimator
Maximum Likelihood	$\hat{\theta}_{ML} = S/n$
Maximum A Posteriori	$\hat{\theta}_{MAP} = \frac{S + \alpha - 1}{n + \alpha + \beta - 2}$
Bayesian posterior mean	$\hat{\theta}_{Bayes} = \frac{S + \alpha}{n + \alpha + \beta}$

1. Determine the bias and variance of these estimators.
2. How does the prior affect bias and variance?
3. Compare the estimators' bias and variance and discuss implications.

Solution (1) — MLE and MAP Estimators

Bias and Variance of MLE: (recall from lecture)

→ $\text{Bias}(\hat{\theta}_{MLE}) = 0, \quad \text{Var}(\hat{\theta}_{MLE}) = \frac{\theta(1-\theta)}{n}$

MAP Estimator:

$$\hat{\theta}_{MAP} = \frac{S + \alpha - 1}{n + \alpha + \beta - 2}$$

$S = \sum_{i=1}^n X_i$

$E[a] = a$
 $E[a \cdot X] = a \cdot E[X]$
 $E[X_1 + X_2] = E[X_1] + E[X_2]$

$$\Rightarrow \text{Bias}(\hat{\theta}_{MAP}) = E[\hat{\theta}_{MAP}] - \theta = E\left[\frac{S + \alpha - 1}{n + \alpha + \beta - 2}\right] - \theta = \frac{E[S] + \alpha - 1}{n + \alpha + \beta - 2} - \theta = \frac{n\theta + \alpha - 1}{n + \alpha + \beta - 2} - \theta$$

$$\text{Var}(\hat{\theta}_{MAP}) = \text{Var}\left(\frac{S + \alpha - 1}{n + \alpha + \beta - 2}\right) = \frac{\text{Var}(S)}{(n + \alpha + \beta - 2)^2} = \frac{m \cdot \text{Var}(X_i)}{(n + \alpha + \beta - 2)^2} = \frac{n\theta(1-\theta)}{(n + \alpha + \beta - 2)^2}$$

$$\text{Var}(S) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = m \cdot \theta(1-\theta)$$

$$\text{Var}(a \cdot X) = a^2 \cdot \text{Var}(X) \quad \longleftrightarrow \quad \text{Var}\left(\frac{S+\alpha-1}{n+\alpha+\beta-2}\right) = \frac{\text{Var}(S+\alpha-1)}{(n+\alpha+\beta-2)^2}$$

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$$

if X_1 is indep. of X_2

For $X \rightarrow \text{Bernoulli}(\theta)$ r.v. we have $E[X] = \theta$
 $\text{Var}(X) = \theta(1-\theta)$ ✓

$$\text{Var}(a) = 0$$

Solution (2) — Bayesian Posterior Mean Estimator

Bayesian estimator (posterior mean):

$$\hat{\theta}_{\text{Bayes}} = \frac{S + \alpha}{n + \alpha + \beta}$$

So:

$$\text{Bias}(\hat{\theta}_{\text{Bayes}}) = E\left[\frac{S + \alpha}{n + \alpha + \beta}\right] - \theta = \frac{E[S] + \alpha}{n + \alpha + \beta} - \theta = \frac{\alpha + n\theta}{n + \alpha + \beta} - \theta,$$

$$\text{Var}(\hat{\theta}_{\text{Bayes}}) = \frac{\text{Var}(S + \alpha)}{(n + \alpha + \beta)^2} = \frac{\text{Var}(S)}{(n + \alpha + \beta)^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{(n + \alpha + \beta)^2} = \frac{n \cdot \theta(1 - \theta)}{(n + \alpha + \beta)^2} = \frac{n\theta(1 - \theta)}{(n + \alpha + \beta)^2}$$

Summary of Results for Question 1

Estimator	Bias	Variance
MLE	0	$\frac{\theta(1-\theta)}{n}$
MAP	$\frac{\alpha + n\theta - 1}{\alpha + \beta + n - 2} - \theta$	$\frac{n\theta(1-\theta)}{(n + \alpha + \beta - 2)^2}$
Bayesian Mean	$\frac{\alpha + n\theta}{\alpha + \beta + n} - \theta$	$\frac{n\theta(1-\theta)}{(n + \alpha + \beta)^2}$

Observations:

- Larger the $\alpha + \beta$ (more informative prior) yields smaller variance but larger bias.
- Compare the variance of Bayesian, MAP, and MLE (see next slide).

Question 1 — Bias and Variance Visualization

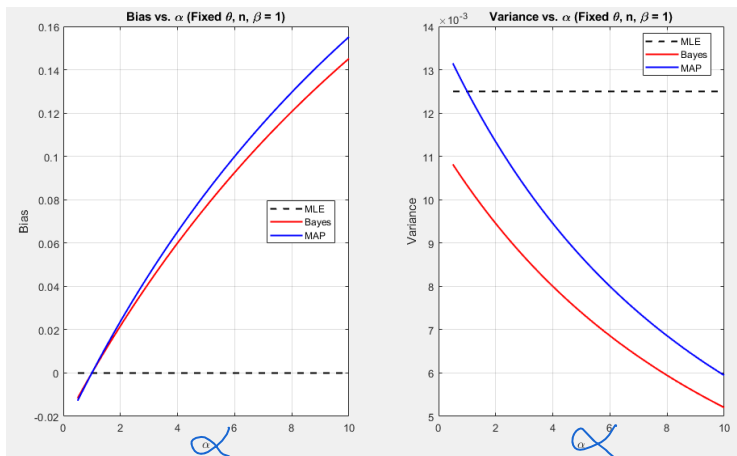


Figure: Effect of prior parameter α on bias and variance, with $\theta = 0.5$ (fair coin), $n = 20$, $\beta = 1$, and varying α .

- ▶ As α increases, variance decreases but bias increases.
- ▶ Illustrates the bias–variance trade-off.

Question 2

Let $X_i \sim \text{Poisson}(\lambda)$, $i = 1, 2, \dots, n$. The PMF of Poisson is:

$$P(X_i = x_i | \lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

- ▶ Derive the MLE and MAP estimators for the Poisson mean λ .
- ▶ Use a $\text{Gamma}(\alpha, \beta)$ prior for the MAP.
- ▶ Compare bias and variance of MLE and MAP.

Note: In closed-book assessment, the PMF of Poisson would be given with the question, and similarly the PDF of the Gamma distribution. In practice exercise it's not, since you can find it for yourself.

Solution (1) — MLE

Let $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. $i = 1, \dots, n$

$$P(X_i = x_i | \lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

PMF

Likelihood function (depends on all X_i 's, but it's a function of λ):

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{X_i} e^{-\lambda}}{X_i!}$$

$$\log L(\lambda) = \sum_{i=1}^n X_i \log \lambda - \lambda \left(\sum_{i=1}^n 1 \right) - \log X_i!$$

Taking log, differentiating and equating to 0 (we've done this before in the Maximum Likelihood exercises):

$$\frac{d \log L}{d \lambda} = \frac{S}{\lambda} - n = 0 \quad \Rightarrow \quad \hat{\lambda}_{MLE} = \frac{S}{n} \quad \checkmark$$

Solution (2) — MAP Estimator with Gamma Prior

Prior: $\lambda \sim \text{Gamma}(\alpha, \beta)$. This is its PDF:

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0$$

$$\prod_{i=1}^S \lambda^{x_i} = \lambda^S$$

Let X denote $X_1, \dots, X_n$. We need to compute the posterior PDF only up to normalising factor (since MAP only needs to find the λ where it achieves its maximum):

$$p(\lambda|X) \propto p(\lambda) \cdot p(X|\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda} \cdot \prod_{i=1}^S \lambda^{x_i} e^{-\lambda} = \lambda^{\alpha+S-1} e^{-(\beta+n)\lambda}$$

$\hookrightarrow$ proportional to (only numerator for Bayes rule)

Taking derivative of log posterior and equating it to 0:

$$\frac{d}{d\lambda} \left[(\alpha + S - 1) \log \lambda - (\beta + n)\lambda \right] = 0$$

Hence:

$$\hat{\lambda}_{MAP} = \frac{\alpha + S - 1}{\beta + n}$$

Solution (3) — Computing Bias and Variance

For $X_i \sim \text{Poisson}(\lambda)$, we know (or we look it up, or we calculate for ourselves):

$$E[X_i] = \lambda, \quad \text{Var}(X_i) = \lambda$$

Bias and variance of MLE:

$$E[\hat{\lambda}_{MLE}] = E\left[\frac{S}{n}\right] = \frac{E[S]}{n} = \frac{n \cdot \lambda}{n} = \lambda, \quad \text{so the bias is } \lambda - \lambda = 0$$

$$\text{Var}(\hat{\lambda}_{MLE}) = \text{Var}\left(\frac{S}{n}\right) = \frac{\text{Var}(S)}{n^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2} = \frac{n \cdot \lambda}{n^2} = \frac{\lambda}{n}$$

Bias and variance of MAP:

$$E[\hat{\lambda}_{MAP}] = \dots\dots\dots = \frac{\alpha + n\lambda - 1}{\beta + n}$$

So,

$$\text{Bias}(\hat{\lambda}_{MAP}) = \frac{\alpha + n\lambda - 1}{\beta + n} - \lambda,$$

$$\text{Var}(\hat{\lambda}_{MAP}) = \dots\dots\dots = \frac{n\lambda}{(\beta + n)^2}$$

Question 2 — Bias and Variance Visualization

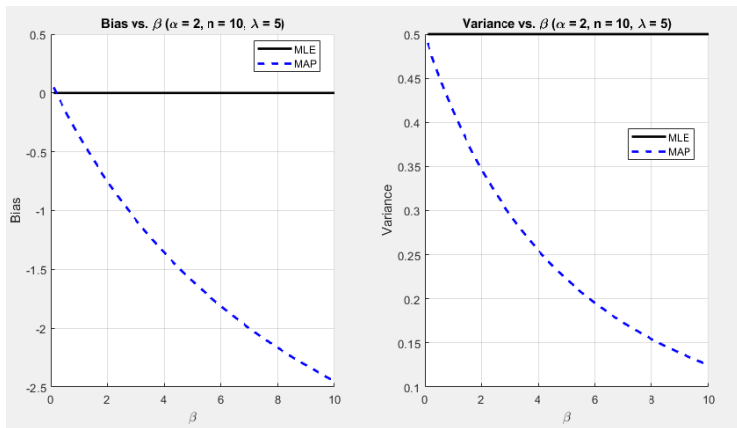


Figure: Bias and variance of the MLE and MAP estimators as β varies.

- ▶ Larger β reduces variance but increases bias.
- ▶ Illustrates the same bias–variance trade-off seen in the Bernoulli case.

Summary of Question 2 Results

Estimator	Bias	Variance
MLE	0	$\frac{\lambda}{n}$
MAP	$\lambda \left(\frac{n}{\beta + n} - 1 \right) + \frac{\alpha - 1}{\beta + n}$	$\frac{n\lambda}{(\beta + n)^2}$

Key takeaway:

- ▶ MAP has smaller variance but introduces bias.
- ▶ Larger β reduces variance further but increases bias.

Question 3 - confidence bounds

1. What could go wrong if a decision tree classifies all training points correctly?
2. How can confidence bounds help choose tree depth?

Solution — Decision Trees

(1) Overfitting:

- ▶ A very deep tree perfectly fits training data but it may generalise poorly to new data.
- ▶ It captures the noise too rather than just the signal.

(2) Using Confidence Bounds:

- ▶ If enough labelled data is available, use **test set bounds**.
- ▶ Otherwise, you can use **Occam's razor bounds** on the training set, especially if you have good bets on the classifiers in the set of classifiers.
- ▶ These provide high-confidence bounds on the true generalisation error.
- ▶ Important in safety-critical applications (e.g. medicine, autonomous driving).

Question 4

a) Answer the following:

- (i) Is zero test error necessarily good?
- (ii) Can the Occam's razor bound justify that it's a good classifier?
- (iii) Given $n = 30$, 4 test errors, find 90% and 95% confidence upper bounds on true error.

b) Model tested on 1000 emails, 850 correct.

- (i) Compute 95% confidence upper bound on true accuracy.
- (ii) Interpret the meaning of confidence interval.

Solution (a) — Discussion and Bounds

(i) Zero test error can be misleading if the test set is small or unrepresentative.

(ii) Occam's razor bound helps justify small true error if:

- ▶ Training sample is i.i.d.
- ▶ Prior bets were made before observing data.

(iii) Test set bound:

$$c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln(1/\delta)}{2n}}$$

For 4 errors out of 30:

$$90\% \text{ bound: } 0.1333 + \sqrt{\frac{\ln(10)}{60}} \approx 0.329$$

$$95\% \text{ bound: } 0.1333 + \sqrt{\frac{\ln(20)}{60}} \approx 0.357$$

Solution (b) — Email Classification Example

Observed error = 0.15 on $n = 1000$ samples.

Using the same bound:

$$c_{\mathcal{D}} \leq 0.15 + \sqrt{\frac{\ln(20)}{2000}} \approx 0.189$$

Interpretation:

- ▶ With 95% confidence, true error ≤ 0.189 .
- ▶ A 95% confidence interval means that across many random test samples, about 95% of such intervals would contain the true accuracy.